Chemical Engineering Science

Machine Learning and Neural Network Interpretation for CO_2 Uptake Prediction in Functionalized Amino Acid Ionic Liquids

--Manuscript Draft--

Manuscript Number:	CES-D-26-06457
Article Type:	Research paper
Section/Category:	Environmental Chemical Engineering
Keywords:	machine learning; neural networks; multiple linear regression; CO_2 absorption; amino acid; functionalized ionic liquid

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- Developed an interpretable machine learning (ML) framework to predict CO_2 absorption in amino acid ionic liquids (AAILs).
- Identified viscosity change and amine functionality as key descriptors governing CO_2 uptake.
- Second-order linear regression achieved high predictive accuracy (R^2 = 0.9957, RMSE = 0.0157) while retaining mechanistic interpretability.
- The proposed ML framework successfully screened and ranked literature-reported AAILs beyond the experimental dataset.

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Shakila Akter [a, b*], Gobinath P. Rajarathnam [a*], Jing-Yi Li [a], Petr Stanovsky [c], Pavel Izak [c, d, e], Wojciech Kujawski [f], Hui-Hsin Tseng [g], Marjorie Valix [a], David K. Wang [a, h*]

[a] *School of Chemical and Biomolecular Engineering, The University of Sydney, New South Wales 2008, Australia*

[b] *Institute of National Analytical Research and Service (INARS), Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhanmondi, Dhaka 1205, Bangladesh*

[c] *Institute of Chemical Process Fundamentals of the CAS, Rozvojová 135, Prague, 16500, Czech Republic*

[d] *Department of Physical Chemistry, University of Chemistry and Technology Prague, Technická 5, 166 28 Prague 6, Czech Republic*

[e] *Institute of Environmental Technology, CEET, VŠB – Technical University of Ostrava 17. listopadu 2172/15, 708 00 Ostrava-Poruba, Czech Republic*

[f] *Faculty of Chemistry, Nicolaus Copernicus University in Toruń, 7 Gagarina Street, Toruń, 87-100, Poland*

[g] *Hierarchical Green-Energy Materials (Hi-GEM) Research Center, National Cheng Kung University, Tainan City 701401, Taiwan*

[h] *Department of Resources Engineering, National Cheng Kung University, Tainan City 701401, Taiwan*

Corresponding authors:
* School of Chemical and Biomolecular Engineering, The University of Sydney, New South Wales 2008, Australia.
Tel.: +61 2 9351 3832; fax: +61 2 9351 2854.
E-mail address: david.wang1@sydney.edu.au
gobinath.rajarathnam@sydney.edu.au

Abstract

The development of efficient amino acid ionic liquids (AAILs) for CO_2 capture is hindered by the large experimental effort required to evaluate the diverse combinations of cation-anion structures and operating conditions. This work presents a machine-learning (ML) framework for predicting the CO_2 absorption capacity of functionalized AAILs using experimentally derived physicochemical descriptors. Three complementary predictive ML models were developed and evaluated: first-order linear regression (FLR), neural networks (NN), and second-order linear regression (SLR). The framework was established from a focused experimental dataset and subsequently assessed using literature-reported AAILs to examine its predictive capability beyond the training data. Among the investigated approaches, the FLR model provided an interpretable baseline (R^2 = 0.8926), while the NN model achieved improved predictive performance (R^2 = 0.9935) by capturing nonlinear relationships. The SLR model delivered the best overall performance (R^2 = 0.9957, RMSE = 0.0157), combining high predictive accuracy with the ability to account for interactions among the governing CO_2 capture parameters. Model analysis consistently identified viscosity evolution during CO_2 absorption, temperature, and amine functionality as the dominant contributors and descriptors for the absorption performance. Application of the developed framework enabled efficient comparative evaluation and prioritization of candidate AAILs reported in the literature without requiring additional experimental measurements. The proposed methodology provides a reliable and computationally efficient strategy for accelerating the assessment of functionalized ionic liquids and supports the broader integration of interpretable machine learning into the development of advanced carbon-capture absorbents.

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- Second-order linear regression achieved high predictive accuracy ($R^2 = 0.9957$, RMSE = 0.0157) while retaining mechanistic interpretability.
- The proposed ML framework successfully screened and ranked literature-reported AAILs beyond the experimental dataset.

1. Introduction

Global warming represents one of the most significant environmental concerns mankind has faced today, and greenhouse gas (GHG) emissions are the primary reason for climate change, with carbon dioxide (CO_2) accounting for approximately 80% of the total emissions.[1] The Intergovernmental Panel on Climate Change (IPCC) investigated and found that the global temperature is expected to increase by around 2 ℃ by the end of the twenty-first century without taking effective action.[2]

To reduce carbon emissions and protect the global environment, it is essential to focus on developing efficient technologies for carbon capture and storage (CCS). Recently, researchers have explored CO_2 absorption processes using environmentally safe, cost-effective green absorbents for capturing carbon to mitigate global heating. Among the developing research, ionic liquids are used as flexible and processable green absorbents with the best potential to introduce an innovative CO_2 capture technology that influences the conventional solvent absorption and membrane processes.[3, 4] Especially, the molecular design of functionalized amino acid ionic liquids (AAILs) with integrated functionalized amino groups (-NH_2) capable of binding CO_2 molecules through their available lone pair electrons to enhance the affinity between the ionic liquid and CO_2 molecule for increasing absorption capacity.[5] For this reason, the adaptable nature of AAILs, aligned with their essential eco-friendliness and low toxicity, makes them a useful and practical absorbent for sustainable carbon capture materials.[6, 7]

Moreover, the structural complexity and diverse chemistries of ILs and AAILs provide rich datasets that can be utilized in the development of advanced predictive models. Machine learning and neural network algorithms are a versatile tool that has the potential to accelerate the discovery of high-efficiency CO_2 absorbents and thus have been increasingly employed to screen and optimize IL structures when trained on carefully curated experimental data. Moosavi et al.[8] applied an artificial-neural-network-based (ANN-based) model to predict the performance of CO_2-foam flooding on a laboratory scale, aiming to improve the oil recovery. In a study by Hamzehie et al.,[9] an artificial neural network was used to forecast the capture of CO_2 and H_2S in 32 distinct types of ionic liquids. A "single decision tree" and an "ensemble random forest" were utilized by Venkatraman et al.[10] to estimate the absorption capacity of CO_2 in different ionic liquids.

Several machine learning studies have reported accurate predictions of CO_2 and H_2S solubility in ionic liquids, primarily by employing extensive physicochemical descriptor sets or detailed molecular properties as model inputs.[11-15] While these models gain the best predictive accuracy, they usually depend on numerous input descriptors (such as temperature, pressure, molecular weight, density, and viscosity) and are fundamentally developed as predictive machine learning tools, offering limited mechanistic insight into the molecular factors governing CO_2 gas absorption. However, the present study applies a concise, chemically explainable descriptor dataset that combines experimentally accessible thermophysical properties with amine functionality to incorporate molecular design and AI-driven machine learning modelling. This approach provides both accurate prediction and mechanistic understanding, enabling the rapid screening and rational design of high-performance ionic liquids for CO_2 capture.

In our study, we present a set of machine-learning modelling approaches for predicting CO_2 uptake in functionalized amino acid ionic liquids (AAILs). Based on a curated experimental dataset, the proposed framework is intended to facilitate comparative evaluation of candidate AAILs and identification of key structure-property relationships of the different amine groups, rather than detailed process-level optimization. Three machine learning (ML) models were developed to predict CO_2 absorption capacity and provide complementary insights: (i) a first-order linear regression (FLR) model including only main effects, (ii) a neural network (NN) model, and (iii) a second-order linear regression (SLR) model incorporating two-way interaction terms.

Crucially, this study demonstrates that pre- and post-absorption viscosities together with primary and secondary amine functionalities are sufficient to accurately predict CO_2 uptake in amino acid ionic liquids. These experimentally accessible descriptors are rarely incorporated explicitly in previous machine learning studies, which typically employ numerous molecular descriptors or graph-based molecular representations to achieve high predictive performance (R^2 values of approximately 0.99). In comparison with the explainable graph neural network developed by Jian et al.,[16] which relies on molecular descriptors derived from ionic liquid structures, our developed models achieve comparable predictive accuracy using a compact set of experimentally measurable thermophysical and structural descriptors. The use of these physically meaningful variables enables direct quantification of the effects of viscosity changes (before and after CO_2 absorption) and amine functionality on CO_2 absorption, providing mechanical perception that complements predictive capability. This corresponding strategy of the ML model generation not only maintains predictive accuracy but also improves the mechanical knowledge of CO_2 capture in AAILs.

2. Methodology

The objective of this study was to identify a suitable modelling framework for predicting CO_2 absorption capacity in amino acid ionic liquids (AAILs). The output variable S corresponds to the measured CO_2 absorption capacity, expressed as mol CO_2/mol IL. It was also modelled as a function of selected molecular and operating descriptors using machine learning techniques, expressed as:

$$S = f\ (\mathrm{I}, \mathrm{T}, \mathrm{P}, \mathrm{V}, \mathrm{A}) \tag{1}$$

These variables were employed as input descriptors for model development, capturing both the intrinsic chemical characteristics of the ionic liquids and the relevant thermodynamic operating conditions.

Where:

I (AAIL type): nominal categorical variable representing the amino acid ionic liquid identity; P[Arg], P[Lys], P[His], P[Gly], P[Br], of which P denotes the trihexyl(tetradecyl)phosphonium cation, and the brackets indicate the respective amino acid anion (AA^-) in the ML models. To ensure meaningful input for the machine learning models without imposing artificial relationships, the AAILs were encoded using a minimal, non-ordinal scheme.

T (Temperature): continuous variable in degrees Celsius (°C), representing the experimental system temperature between 15 and 40 °C during CO_2 absorption.

P (Pressure, $\Delta p(t)$): continuous variable treated as a scalar corresponding to the total change in CO_2 partial pressure (bar) over the absorption period. Although pressure is measured as a function of time, temporal information is not explicitly modelled, as the objective of this study is to predict overall CO_2 uptake for comparative screening rather than absorption kinetics.

V (Viscosity): continuous variable, measured in Pa·s, recorded both pre and post CO_2 absorption and the percentage change in viscosity was also computed in the model. Although these variables are correlated, all three were retained to represent baseline fluid properties, viscosity under CO_2 loading, and the relative rheological response to absorption. Diagnostic checks confirmed that their inclusion did not compromise model stability or predictive performance.

A (Amine group functionality): continuous variable representing the quantity of active amine groups (0, 1, 2, etc.) was encoded in the ionic liquid. It showed that the number of primary (-NH_2) and secondary (-NH) amines was operated independently and assigned separate contributions to the descriptor, reflecting their different reactive sites toward CO_2 molecules. This encoding allows the model to distinguish between ionic liquids with the same total amine count but different amine group types.

2.1. Data acquisition of the AAILs dataset

For developing the machine learning (ML) models and multiple linear regression (MLR), we used the JMP Pro 18 software least squares technique [17] for data generation from the original experimental dataset of the CO_2 absorption in AAILs. The 60 data points in the examined dataset represent CO_2 absorption capacity measured with multiple measurements for each ionic liquid under various pressure and temperature conditions (15-40 °C). These data showed five functionalized ionic liquids: arginate [Arg], lysinate [Lys], histidinate [His], glycinate [Gly], and bromide [Br] with a phosphonium cation (P^+) but differ in the anion species (AA^-). Each data point includes the number and type of amine functionalities, temperature, pressure, viscosity before and after CO_2 absorption, and the corresponding CO_2 uptake. The measured absorption capacities range from 0.05 to 0.72 mol CO_2/ mol IL. All observations for model building were consistent with experimental uncertainties and physically meaningful patterns, with no outliers eliminated.

Table 1. Statistical parameter estimates of the FLR model.

Descriptor	Parameter	Value	Standard Error	t Ratio	Prob > |t|
Intercept	w_0	0.2399086	0.096889	2.48	0.0166*
Temperature (℃)	w_1	-0.003853	0.001883	-2.05	0.0459*
Pressure (bar)	w_2	0.0364779	0.037858	0.96	0.3397
Viscosity (before CO_2 absorption)	w_3	-0.449897	0.336714	-1.34	0.1873
Viscosity (after CO_2 absorption)	w_4	0.3468493	0.354167	0.98	0.3319
Viscosity (% Change)	w_5	0.0016412	0.000615	2.67	0.0101*
Primary (1°) Amine	w_6	0.0276611	0.01813	1.53	0.1331
Secondary (2°) Amine	w_7	0.0121903	0.012007	1.02	0.3147

Note: * indicates statistically significant coefficients at $p < 0.05$ (95% confidence level).

2.2. First-order linear regression (FLR) model

The FLR model was built under the standard regression interpretations: (i) a linear relationship between descriptors and CO_2 absorption, (ii) independence of data points, and (iii) absence of strong multicollinearity among descriptors. To estimate the parameters of the model using the ordinary least squares (OLS) method, which minimises the inconsistency between experimental and predicted values. The model was developed to analyse the CO_2 absorption capacity in AAILs as a function of significant physicochemical parameters, including temperature, pressure, primary and secondary amine groups, viscosity (before and after CO_2 absorption), and the percentage change in viscosity. This approach influences the most important characteristics that control the CO_2 absorption parameter, which allows for yielding interpretable coefficients of the model.

The specific descriptors and parameter values obtained from the fitted model are summarized in Table 1. In the Linear Regression (LR) method, linear equations are regressed between data and independent variables, where simplicity is one of its benefits. The FLR model has been developed for predicting CO_2 absorption in AAILs, as shown in Eq. (2), and this approach was taken since it significantly improved the accuracy of the LR model.

$$y = w_0 + w_1\beta_1 + w_2\beta_2 + \cdots + w_i\beta_i \quad (2)$$

Where y denotes the CO_2 absorption capacity (mol CO_2/mol IL), w_i are regression coefficients, and β_i are the input descriptors. Standard errors and 95% confidence intervals for the FLR model coefficients were obtained using least-squares regression inference, based on the estimated residual variance and the Student's t-distribution, which is reported in the supporting information (Figure S1). The magnitude and the sign of the coefficients, together with their confidence intervals, provide quantitative insight into the relative influence of each descriptor on CO_2 absorption.

2.3. Neural Networks (NN) model

Neural networks in machine learning are computational models inspired by the structure and function of the biological nervous system. In this work, a neural network (NN) model was developed to predict CO_2 absorption capacity (mol CO_2/mol IL) from ionic liquid and amino acid-based system parameters. Predictor variables included amino acid IL (AAIL), temperature (°C), pressure (bar), viscosity before and after CO_2 absorption, viscosity percentage change, and number of primary (1°) and secondary (2°) amines. The model used a single hidden layer with three TanH nodes and three Gaussian nodes. This combination of activation functions was chosen to capture both non-linear interactions (TanH, Gaussian) effects, thereby improving model flexibility. The NN model summary is included in the supporting information (Figure S2).

Model training employed 5-fold cross-validation (k-Fold =5), with data partitioned at the ionic liquid level to ensure that measurements corresponding to the same AAIL chemistry were not distributed across training (experimental) and validation folds. To ensure repeatability, a fixed random seed (1) was used and all continuous input descriptors were standardised before model training. A squared penalty and one fitting epoch were used to execute regularisation and for model training because more epochs did not improve model performance given the size of the current experimental dataset. So, the performance of the model was evaluated by comparing predicted CO_2 absorption values with experimental data.

2.4. Second-order linear regression (SLR) model

The second-order linear regression (SLR) model was developed to quantify both the main effects of physicochemical descriptors and their pairwise interactions on CO_2 absorption. The model focused only on two-way interaction effects, which allows for the detection of synergistic or antagonistic relationships between variables while maintaining a relatively simple and interpretable model structure. Regression-based approaches of this type have been widely applied to predict CO_2 solubility in ionic liquids.[18] Model coefficients were estimated using ordinary least squares (OLS), and model stability was ensured through systematic screening of interaction terms based on statistical significance and multicollinearity diagnostics prior to final model selection, as reported in the supporting information (Figure S3). Cross-validation was employed to evaluate predictive robustness and confirm that inclusion of interaction terms did not lead to overfitting. The general form of the SLR model is given by Eq. 3:

$$y = \beta_0 + \sum_{i=1}^{n} \beta_i X_i + \sum_{i<j}^{n} \beta_{ij} X_i X_j + \in \quad (3)$$

where y denotes the predicted CO_2 absorption (mol CO_2/mol IL), X_i is the molecular or process descriptors, β_0 is the intercept, β_i are main effect coefficients, β_{ij} are coefficients for two-way interaction terms, and $\in$ is the error term.

3. Results and discussion

3.1. CO_2 absorption via the FLR model

Linear Regression (LR) is a widely used ML method for modelling IL properties such as density, viscosity, surface tension, and melting point temperature.[19-21] In addition, multiple linear regression (MLR) has been applied to modelling CO_2 solubility in ILs.[22] As the starting point of our modelling framework, we developed a first-order linear regression (FLR) model to establish an interpretable baseline for predicting CO_2 absorption in amino acid ionic liquids (AAILs). In this study, the FLR model was used on an original experimental and focused dataset of CO_2 absorption in AAIL systems,[23] where linear models are less prone to overfitting than more complex machine learning approaches and thus provide a stable benchmark for evaluating advanced models.[24]

In our previous study, we created and validated a multiple linear regression (MLR) model, which showed consistent predictive performance for calculating the CO_2 absorption capacity of AAILs.[23] The model explained above 90% of the overall variance in the dataset with a high R^2 value and had a low root mean square error (RMSE = 0.06) value, demonstrating good agreement and a strong correlation between the predicted and measured CO_2 absorption capacities. The highly significant statistical result ($p < 0.0001$) demonstrated the model's reliability and ability to pinpoint the key factors influencing CO_2 capture performance. In this work, residual analysis is used to further assess the FLR model as a baseline prediction strategy within the machine learning framework, and the summary of the model is included in Figure S1 in the supporting information.

The residual versus predicted plot (Figure 1) shows that the residuals are randomly distributed about the zero-reference line (blue) across the full range of predicted CO_2 absorption capacities. The FLR model does not regularly overpredict or underpredict the experimental data, as evidenced by the random distribution of residuals around this line, which also confirms the adequacy of the regression model and shows the lack of systematic prediction bias. Additionally, it appears that constant error variance is satisfied based on the comparatively uniform spread of the residual analysis. Overall, these findings validate the linear regression model's suitability for characterising the key variables controlling CO_2 absorption in the examined AAILs. The residual distribution shows discernible variability in the intermediate prediction region, where both positive and negative deviations from zero are seen, despite this usually acceptable behaviour.

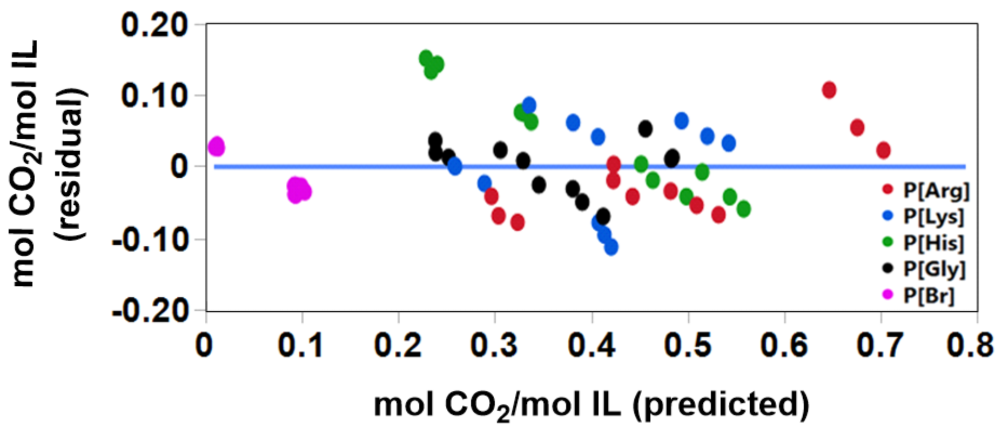


Figure 1. Residual vs predicted CO_2 absorption in AAILs of the FLR model. The blue horizontal line indicates the zero-residual reference (Residual = 0).

The residual distribution exhibits noticeable variability in the intermediate prediction region, where both positive and negative deviations from zero are observed, which is usually acceptable behaviour. This spread demonstrates the limited capacity of the FLR model to represent complex connections between process variables, particularly those relating to contributions from amine functionality, temperature effects, and changes in viscosity during absorption. These differences suggest that there may be more nonlinear linkages influencing the CO_2 absorption process than a strictly linear explanation can account for.

Table 2. Statistical effect test of the FLR model.

Source	Sum of Squares	F Ratio	Prob > F
Temperature (°C)	0.01594986	4.1852	0.0459*
Pressure (bar)	0.00353819	0.9284	0.3397
Viscosity (before CO_2 absorption)	0.00680367	1.7853	0.1873
Viscosity (after CO_2 absorption)	0.00365516	0.9591	0.3319
Viscosity (% Change)	0.02718408	7.1330	0.0101*
Primary (1°) Amine	0.00887129	2.3278	0.1331
Secondary (2°) Amine	0.00392794	1.0307	0.3147

Note: * indicates statistically significant coefficients at $p < 0.05$ (95% confidence level).

The residual magnitude exhibits small shifting patterns with higher anticipated absorption values, suggesting growing forecast error in this area. This behaviour could be explained by the increased sensitivity of CO_2 absorption performance at higher capacities, where more complex physicochemical interactions occur. Further supporting these findings, the statistical output linked with the model is shown in Table 2. As shown by the substantial statistical contributions, temperature and viscosity changes have a major impact on controlling the absorption behaviour of the predicted CO_2 capacity. On the other hand, pressure, viscosity before and after absorption, and number of amine groups exhibit much weaker statistical influence, suggesting that these effects are less noticeable inside the first-order linear framework.

3.2. CO_2 absorption via the NN model

Overall, the residual features confirm that the FLR model provides a reasonable first-level estimation of CO_2 absorption capacity in AAIL systems as a reliable baseline model. However, the localised prediction variability and residual dispersion show that the linear approach is unable to adequately capture the intrinsic complexity of the absorption mechanism. These limitations inspire the use of more sophisticated modelling techniques, such as a neural network (NN) model, which is anticipated to enhance prediction performance and more accurately describe nonlinear interactions. Therefore, neural networks (NN) can be applied to complex modelling tasks because they fit parameters jointly across multiple descriptors, allowing them to capture nonlinear relationships.[25] NN models have been widely developed in predicting ionic liquid (IL) properties,[19, 26] and CO_2 solubility in ILs by using GC[27] and PaDel[28] descriptors. However, NNs can also be vulnerable to overfitting, where the model learns noise in addition to true patterns in the data.[21]

Building on these advances and in response to the limitations identified in the FLR model, we developed an NN model to capture nonlinearities and hidden interactions that cannot be resolved by linear approaches. The decision to employ an NN framework was motivated by the need to improve predictive accuracy by explicitly accounting for nonlinear coupling between thermodynamic and molecular descriptors of AAILs in relation to CO_2 absorption.

Table 3. Training and validation datasets of the NN model.

Training Performance		Validation Performance	
Measures	**Value**	**Measures**	**Value**
RSquare	0.9934743	RSquare	0.9880608
RASE	0.0228132	RASE	0.0221043
Mean Abs Dev	0.0111311	Mean Abs Dev	0.0157447
-LogLikelyhood	-134.6336	-LogLikelyhood	-29.49276
SSE	0.0249812	SSE	0.0058632
Sum Freq	48	Sum Freq	12

Figure 2 shows the NN model that was developed for this investigation. Eight descriptors made up the input layer. The network was able to identify non-additive patterns in the data by processing these inputs through a hidden layer that included nonlinear activation functions (three Tanh and three Gaussian nodes). The output layer generates predictions of CO_2 absorption capacity (mol CO_2/mol IL) after the hidden nodes combine weighted contributions from all descriptors. This NN diagram enables the model to better capture interactions between chemical functionality and transport parameters, potentially leading to a deeper mechanistic understanding of the dual chemical and physical character of CO_2 uptake in AAILs.

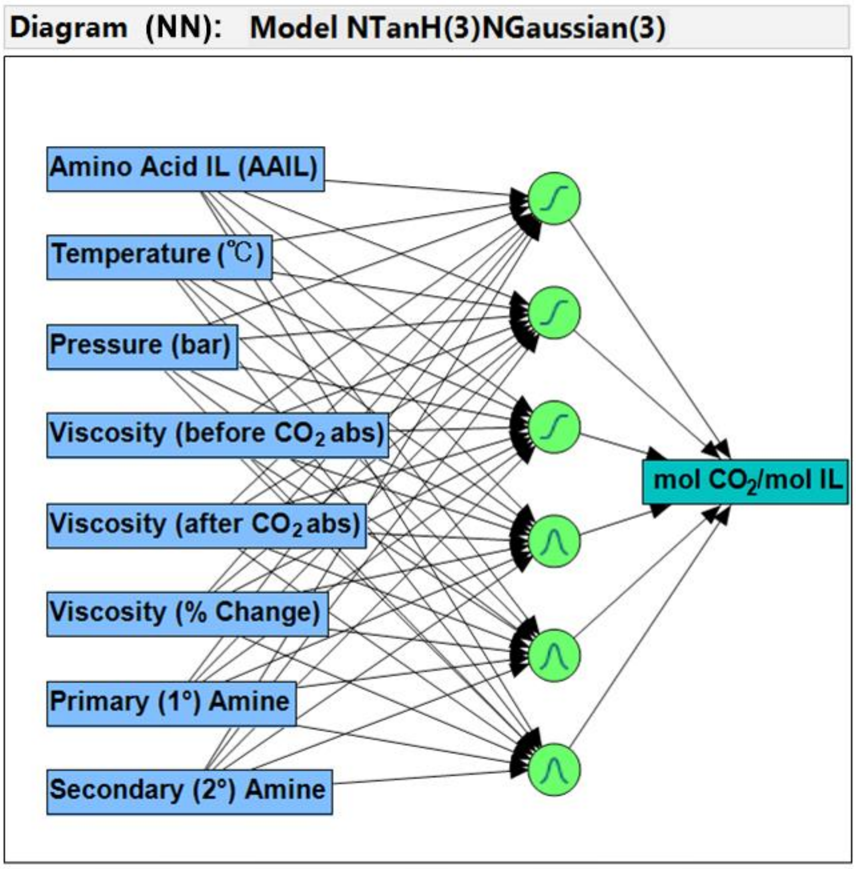


Figure 2. Neural network (NN) diagram for predicting CO_2 absorption in AAILs.

Moreover, Figure 3 shows the created NN model's prediction strength, while Table 3 summarises the statistical information. The anticipated and experimental CO_2 absorption capabilities in AAILs were found to be in great agreement with NN modelling. While validation on an independent dataset (n = 12) yielded similarly high results (R^2 = 0.9881, RASE = 0.0221, MAD = 0.0157), the NN achieved R^2 = 0.9935, RASE = 0.0228, and MAD = 0.0111 on the training dataset (n = 48). The robustness of the model was confirmed by the residual sum of squares remaining low for both training (SSE = 0.0249) and validation (SSE = 0.0059). All of these findings show that the NN model provides near-perfect predictive accuracy for both the training and validation sets, effectively capturing the essential physicochemical patterns controlling CO_2 absorption in AAILs.

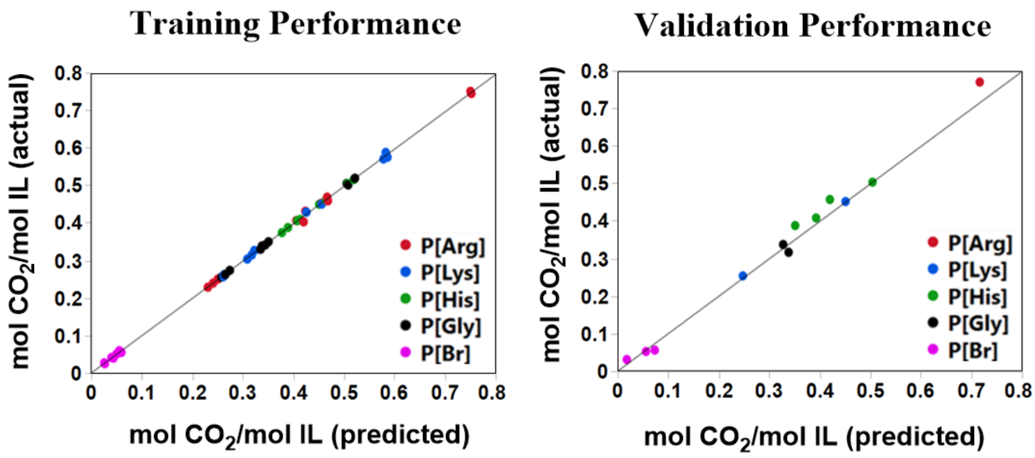


Figure 3. Training and validation performance of the NN model for CO_2 absorption in AAILs.

Nevertheless, the NN model outperforms the FLR model in terms of prediction, but it is a "black box" with poor interpretability, making it challenging to identify and measure the contribution of specific descriptor interactions. The model does a good job of capturing nonlinearities and hidden dependencies between descriptors, but it is unable to identify the factors that are principally in charge of increased CO_2 absorption or how these relationships affect CO_2 uptake. Because of this restriction, it is less useful for rational design and mechanical comprehension. We extended our ML framework to a second-order linear regression (SLR) model, which specifically includes two-way interaction between descriptors, in order to minimize this gap to increase interpretability for producing a more consistent fit for CO_2 absorption capacity in AAIL systems.

3.3. CO_2 absorption via the SLR model

To advance beyond the limitations of the first-order regression and the interpretability gap of the neural network, we constructed a second-order linear regression (SLR) model that explicitly accounts for descriptor-descriptor interactions. This framework was intended not only to improve predictive reliability but also to clarify which combinations of thermodynamic and molecular descriptors most strongly drive CO_2 capture in AAIL systems.

Table 4. Statistical effect summary of the SLR model.

Source	Logworth	PValue
Primary (1°) Amine*Secondary (2°) Amine	5.721	0.00000
Temperature (°C)*Viscosity (after CO_2 abs)	4.678	0.00002
Temperature (°C)*Viscosity (% Change)	3.227	0.00059
Temperature (°C)*Viscosity (before CO_2 abs)	2.964	0.00109
Primary (1°) Amine	2.179	0.00662
Viscosity (after CO_2 abs)*Viscosity (% Change)	1.834	0.01464
Viscosity (after CO_2 abs)*Secondary (2°) Amine	1.715	0.01929
Pressure (bar)*Primary (1°) Amine	1.688	0.02052
Temperature (°C)*Primary (1°) Amine	1.594	0.02550
Viscosity (before CO_2 abs)*Primary (1°) Amine	1.498	0.03180
Viscosity (% Change)*Primary (1°) Amine	1.066	0.08591
Pressure (bar)*Viscosity (after CO_2 abs)	0.989	0.10261
Temperature (°C)	0.688	0.20512
Pressure (bar)*Viscosity (before CO_2 abs)	0.542	0.28684
Pressure (bar)	0.536	0.29110
Viscosity (% Change) * Secondary (2°) Amine	0.522	0.30074
Temperature (°C) * Secondary (2°) Amine	0.518	0.30369
Pressure (bar) * Secondary (2°) Amine	0.350	0.44645
Pressure (bar)*Viscosity (% Change)	0.171	0.67377
Secondary (2°) Amine	0.084	0.82507
Temperature (°C)*Pressure (bar)	0.045	0.90172
Viscosity (before CO_2 abs)*Secondary (2°) Amine	0.032	0.92925
Viscosity (after CO_2 abs)*Primary (1°) Amine	0.022	0.95098
Viscosity (before CO_2 abs)*Viscosity (after CO_2 abs)	0.003	0.99295

Figure 4 demonstrates how well the SLR model predicts CO_2 uptake, with experimental and projected values closely aligned ($R_2 = 0.9957$, $R^2\_adj = 0.9921$, $p < 0.0001$). The small 95% confidence band (shaded red) shows great precision and low uncertainty of the model fit, while the fitted regression line (red) closely tracks the parity between anticipated and actual values. The SLR model has far better prediction capabilities than the mean-response estimate, as seen by the horizontal blue line that represents the dataset's mean experimental CO_2 absorption and acts as a reference baseline. The correctness and precision of the model fit are highlighted by the low error value (RMSE = 0.0157), which shows that adding second-order descriptor interaction significantly enhances performance in comparison to the FLR model's simple linear regression. This outcome proves that interaction effects between descriptors are essential for accurately predicting the CO_2 absorption behaviour of AAILs.

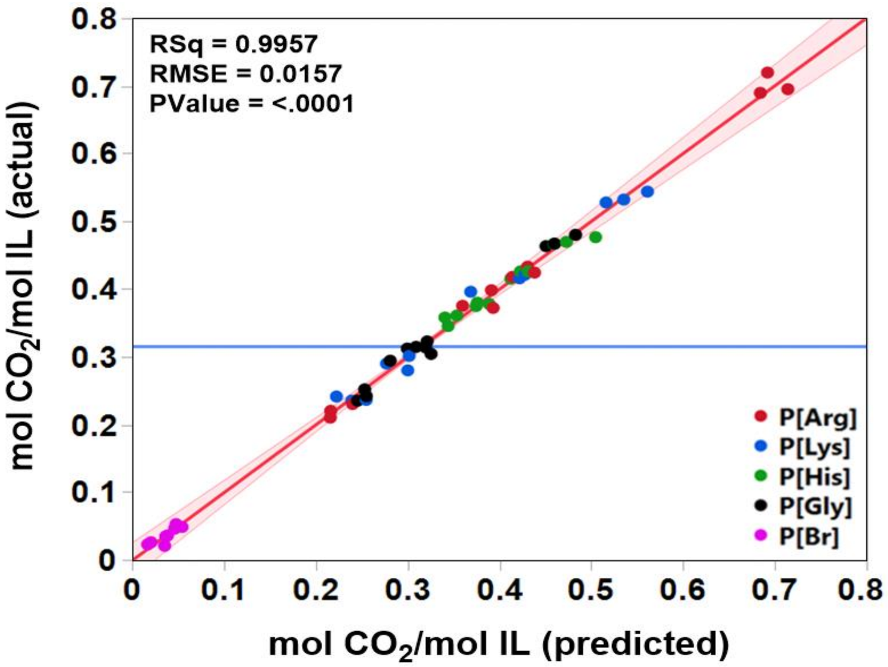


Figure 4. Actual vs predicted CO_2 absorption in AAILs of the SLR model. The red line represents the fitted regression, the shaded red region denotes the 95% confidence interval of the fit, and the blue horizontal line indicates the mean experimental CO_2 absorption (mean response) of the dataset.

Furthermore, the statistical analysis of the SLR model shows the key contributors to the CO_2 absorption performance, as summarized in Table 4. The model was trained using $n = 60$ experimental data points for validation. Primary and secondary amine functionality interaction was the most significant factor (logworth = 5.721, $p < 0.00000$), highlighting the cooperative role of chemical functions in promoting CO_2 binding. Temperature and viscosity after absorption (logworth = 4.678, $p = 0.00002$), temperature and viscosity change (logworth = 3.227, $p = 0.00059$), and temperature and viscosity before absorption (logworth = 2.964, p =

0.00109) were additional significant interacting descriptors. Another significant individual predictor was primary amine concentration (logworth = 2.179, $p = 0.00662$). On the other hand, temperature, pressure, and secondary amine concentration had little explanatory value when taken separately ($p > 0.20$). Additionally, the overall significance of the model was confirmed through ANOVA ($F = 275.60$, $p < 0.0001$), indicating that the SLR model consistently demonstrates the structural and physicochemical factors influencing CO_2 absorption.

Unlike the neural network (NN), which gives accurate predictions without showing why, the SLR model clearly identifies and measures how different descriptors interact. This combination of good predictive performance and clear interpretation makes the SLR the most useful model for understanding the real factors that enhance CO_2 absorption in functionalized AAILs while retaining mechanistic interpretability, offering a transparent alternative to conventional NN 'black-box' machine learning approaches.

3.4. Comparative evaluation of ML models

The sequential modelling strategy highlights how predictive performance and interpretability develop across the three model approaches. Although temperature and viscosity change were identified as significant descriptors by the FLR model ($R^2 = 0.8926$, RMSE = 0.0617), its restricted fit demonstrated that absorption behaviour cannot be adequately described by single-parameter correlations. The NN model produced the best numerical performance ($R^2 = 0.9935$, RASE = 0.0228), with the very high fit may be overstated given the focused dataset and correlations across descriptors, which demonstrated its capacity to capture nonlinearities, but harder to determine the real mechanical factors influencing CO_2 absorption. By explicitly including two-way interaction with descriptors, the SLR model maintained statistical interpretability while achieving good predictive accuracy ($R^2 = 0.9957$, RMSE = 0.0157).

Therefore, the SLR model avoids the potential overfitting risks of a black-box model trained on a modest and correlated dataset compared to the NN model. This balance of accuracy with mechanistic clarity makes the SLR model outputs more reliable and informative, providing confidence that the identified descriptors reflect genuine drivers of CO_2 absorption in AAILs rather than generated of model complexity. Table 5 summarizes the performance of the developed models in comparison with literature-reported models for CO_2 absorption in ionic liquids.

Table 5. Comparison of developed models with literature-reported models for CO_2 absorption using ionic liquids.

Model Types	Descriptors	Reported Performance	References
FLR (single effects)	Process and functionality (e.g., viscosity changes, T, P, and amine groups) parameters	R^2 = 0.893 RMSE = 0.062	This work
NN	Same parameters as above	R^2 = 0.994 RASE = 0.023	This work
SLR (two-way interactions)	Curvature and pairwise with parameters	R^2 = 0.996 RMSE = 0.016	This work
GNN / fingerprint-ML	Graph/fingerprint	R^2 = 0.988, MAE = 0.014	[16]
ANN / LSTM	53 features (incl. T, P, functional groups)	R^2 = 0.986 (ANN test), RMSE = 0.027; R^2 = 0.985 (LSTM test)	[29, 30]
MLR / ANN / RF	σ-profile (COSMO-RS)	R^2 = 0.975, MAE = 0.026 (RF)	[31]
Quadratic/RSM Process vars (e.g., T, P)	Process vars (e.g., T, P)	Polynomial surfaces are widely used; RSM and ML for ILs	[32, 33]
MLR (GA-selected) vs LS-SVM	Structure-based descriptors	Non-linear > linear; MLR as interpretable baseline	[34]

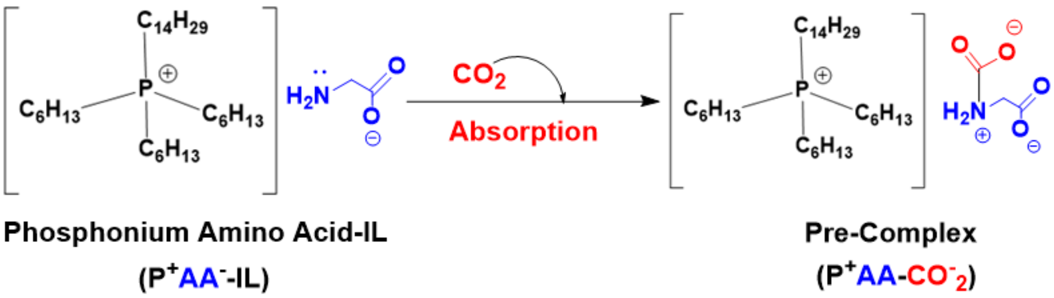
Recent research has shown that fingerprint-based models and graph neural networks are useful for predicting CO_2 absorption in ionic liquids. These models provide insight into substructural contributions and achieve great accuracy with R^2 values of approximately 0.988.[16] Similar to this, deep learning models based on Artificial Neural Network (ANN) and Long Short-Term Memory (LSTM) and also three ML models, Gaussian Process Regression (GPR), LightGBM, and CatBoost, with three descriptors of group contribution (GC), molecular structure descriptors (MSD), and hybrid GC-MSD [29, 30] that were trained on sizable datasets of roughly 10,000 experimental points have demonstrated strong predictive performance and, via sensitivity analysis, have identified important drivers like temperature, pressure, and functional group chemistry. With R^2 values of approximately 0.975, structure-based methods utilising COSMO-RS σ-profile models have also demonstrated remarkable accuracy, underscoring the importance of molecular descriptors for property prediction.[31]

Within this context, it should be noted that direct numerical comparison across literature studies is inherently limited, as the reported metrics are derived from different datasets, descriptor sets, and experimental ranges. In particular, RMSE and MAE values depend strongly on the size, variability, and noise of the underlying data, as well as on whether models are trained on process conditions, structural descriptors, or a combination of both.

3.5. Chemistry explanation of CO_2 absorption

CO_2 absorption is controlled by interdependent chemical and physical processes with synergistic effects across amine functionality, viscosity, and operating temperature rather than by individual descriptors influenced by the absorption chemistry of the AAIL system.

The SLR model shows the balance between reactivity (amine group types), thermodynamics (temperature), and transport (viscosity), providing a mechanistically grounded framework for the rational design of next-generation ionic liquids for carbon capture. The molecular structures and the CO_2 absorption mechanism in AAILs are provided in Scheme 1. The inclusion of data for the second-order with two-way interactions of descriptors in the SLR model not only improved predictive accuracy but also revealed clear chemical rationale effects:



Scheme 1. CO_2 absorption mechanism in AAILs.

Primary (1°) amine vs secondary (2°) amine:

CO_2 uptake is increased when primary and secondary amine sites coexist, with each site functioning independently. The primary amine provides a nucleophilic site for the formation of carbamates, while a neighbouring secondary amine may stabilise intermediates, aid in proton transfers, or act as a base for deprotonation processes.[35] The combination of two amine groups increases CO2 absorption efficiency beyond the sum of their individual effects.

Temperature vs viscosity interactions:

The effect of temperature on CO_2 uptake is highly dependent on the viscosity increase that occurs after carbamate formation. When post-absorption viscosity is high, diffusion becomes rate-limiting, and elevated temperatures disproportionately enhance CO_2 transport and reaction rates.[36] By coupling temperature with viscosity descriptors, the SLR model reflects the reaction of the CO_2 transport trade-off inherent to AAIL-based systems.

Viscosity before vs after CO_2 absorption:

The formation of carbamate or bicarbonate species, which improve ion-ion interactions inside AAILs and the CO_2 molecule, is reflected in an increase in viscosity following CO_2 absorption. Viscosity change is a clear sign of the degree of chemisorption since this structural rearrangement limits ion mobility. So, the SLR model recognises this percentage change as predictive, offering a trustworthy explanation for CO_2 absorption.

Primary Amine (1°) functionality:

The quantity and accessibility of primary amine (-NH_2) groups are directly impacted for the CO_2 capture ability. The amino group interacts with the CO_2 molecule as a nucleophile to produce a stable pre-complex (carbamate) species.[37] Among all AAILs, P[Arg] and P[Lys] have more primary amines, exhibit higher CO_2 absorption, and larger viscosity increases post-absorption, consistent with extensive pre-complex formation as shown in Scheme 1.

Pressure vs primary amines:

The pressure of CO_2 gas directly controls the amount of physically dissolved CO_2 in the ionic liquid; however, the significance of this effect depends on the presence of reactive primary amines. Dissolved CO_2 is quickly transformed into carbamate in systems with a high concentration of -NH_2 groups, which effectively lowers the free CO_2 concentration and maintains the driving force for more CO_2 absorption.

3.6. Simulation of CO_2 absorption with experimental and literature data

CO_2 absorption was simulated for a combined dataset of 421 AAIL systems, comprising 361 literature-reported AAILs with various cationic structures and 60 experimentally synthesised phosphonium-based AAILs, in order to test the predictive accuracy of the SLR model and the generalisability of the developed ML framework.[38-66] Strong predictive accuracy and statistical significance were demonstrated by the model's high coefficient of determination ($R^2 = 0.9669$) and low error value of RMSE of 0.057 ($p < 0.0001$), as shown in Figure 5.

Additionally, most experimental and literature-derived AAIL data points are concentrated around the fitted regression line, while the narrow (95%) confidence envelope reflects the high reliability of the predictions throughout the investigated range. The horizontal reference line denotes the average experimental CO_2 absorption of the dataset, highlighting that the developed SLR model simulation of CO_2 absorption substantially improves prediction beyond a simple mean-response approximation. These findings show how well the simulation model captures CO_2 absorption behaviour in both structurally varied systems reported in the literature and synthesised phosphonium AAILs. The SLR model effectively reflects the basic links between operating circumstances and CO_2 absorption behaviour, as evidenced by this tight agreement. The robustness and transferability of the model are confirmed by the precise prediction of both experimental and literature-reported systems, underscoring its promise as a trustworthy screening tool for assessing and creating high-performance AAILs for CO_2 capture.

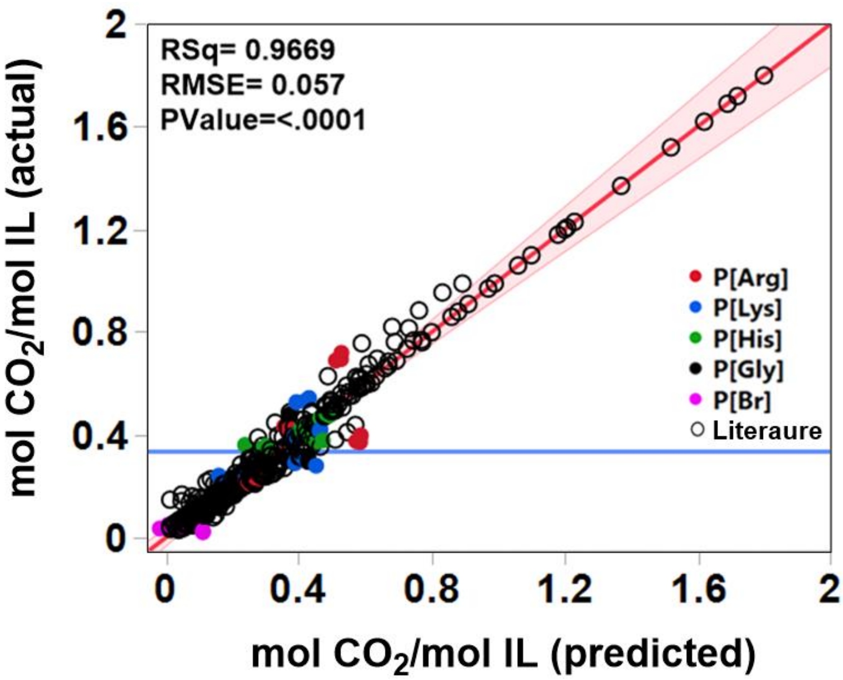


Figure 5. Simulating CO_2 absorption for the experimental and literature-reported AAILs dataset by using the SLR model. The red line represents the fitted regression, the shaded red region denotes the 95% confidence interval of the fit, and the blue horizontal line indicates the mean experimental CO_2 absorption (mean response) of the dataset.

3.6.1. Predictive screening and interpretation of AAILs

The developed SLR model simulation of CO_2 absorption of the literature-reported AAILs provides a rapid and interpretable framework for screening AAILs and identifying the key factors governing CO_2 absorption. The effect summary of the simulation (Figure S4) reveals that temperature is the most influential parameter, followed by amino acid identity and its interaction with pressure, whereas pressure alone exerts a comparatively smaller effect. This suggests that the physicochemical properties of the amino acid anion and how they react to operating conditions are the main factors influencing CO_2 absorption.

The relative performance of the experimental AAILs is further differentiated by the screening outcomes. Due to the increased availability of reactive amine functionalities that promote CO_2 binding, P[Arg] and P[Lys] consistently showed the highest CO_2 absorption capabilities among the synthesised systems. On the other hand, P[His], P[Gly], and P[Br] showed relatively lesser absorption, indicating fewer advantageous interactions with CO_2 molecule.

It is worth mentioning that the literature-reported AAILs exhibiting the highest predicted CO_2 absorption capacities (> 0.8 mol CO_2/mol IL) were generally based on amino acid anions with multiple or readily accessible primary amine groups, particularly similar-derived anions, although they were paired with cations different from the phosphonium cation investigated in this study. This observation is consistent with the simulation developed SLR model, which identified amine functionality, viscosity-related properties, and temperature as the dominant descriptors governing CO_2 uptake, resulting in excellent predictive performance (R^2 = 0.9969). Even though these literature-derived AAILs showed remarkable CO_2 absorption capacities, the majority of earlier research concentrated mostly on equilibrium uptake and offered no details on viscosity changes during absorption. Our current model provides a more substantial mechanistic understanding and a more dependable foundation for the quick screening and logical design of next-generation AAILs for CO_2 capture by directly combining viscosity characteristics with amine functionality.

4. Conclusion

This study introduces an AAIL-specific, viscosity-aware machine learning framework for predicting CO_2 absorption, addressing a class of ionic liquids that has received comparatively limited attention in prior modelling studies despite their unique promise for carbon capture. By explicitly incorporating pre- and post-absorption viscosity together with amine functionality as mechanistic descriptors, our models move beyond conventional screening approaches and capture the structural-transport interplay that governs CO_2 uptake in amino acid ionic liquids.

Using three complementary models of first-order linear regression (FLR), neural networks (NN), and second-order linear regression (SLR), we demonstrated that data-driven methods can generalize from curated experimental datasets to a wide range of literature-reported AAILs, enabling predictive screening and rank-ordering of new candidates in this family of emerging green absorbents. The FLR model (R^2 = 0.8926) established a transparent baseline and identified viscosity changes, temperature, and primary amine as key determinants. The NN model (R^2 = 0.9935) achieved near-perfect predictive accuracy by capturing nonlinear couplings, but at the cost of interpretability. The most balanced result was provided by the SLR model (R^2 = 0.9957), which incorporates interaction terms into a regression framework. It explicitly quantified the roles of viscosity changes and amine group functioning while maintaining mechanistic clarity and near-NN model accuracy.

The SLR coefficients are analysed to provide clear design implications for high-performance AAILs. An important indicator of the intimate connection between chemisorption and gas transport characteristics was the change in viscosity during CO_2 absorption. Although good CO_2 binding is reflected in viscosity increases, excessive viscosity rise may restrict mass transfer. Therefore, to maintain effective gas transmission, the ideal AAIL design should strike a balance between strong CO_2 absorption and appropriate viscosity changes. Additionally, amine functionality has a predictable role, with primary amine groups having a more favourable impact on absorption than secondary amine groups, which is consistent with more advantageous CO_2 binding reaction pathways. Furthermore, validation against simulated, previously reported AAILs confirmed that predicted trends align with experimental behaviour during CO_2 uptake, demonstrating the robustness of the SLR model within the trained chemical domain. Overall, this work shows that interpretable machine learning can both accelerate AAIL screening and provide chemically meaningful guidance for the rational design of next-generation CO_2 absorbents.

5. Data and Software Availability

The datasets used for machine learning model development and validation are provided as Supporting Information in machine-readable Microsoft Excel format. The Supporting Information includes the experimental dataset, the literature-reported dataset, and the machine learning processed dataset used in this study. Machine learning analyses were performed using JMP® Pro 18.2.1.

6. Acknowledgement

D. K. Wang gratefully acknowledges the financial support given by the Australian Research Council Discovery Project (DP190101734), Australia; D. K. Wang, J-Y. Li, H-H. Tseng acknowledges the National Science and Technology Council of Taiwan, ROC (Taiwan Dragon Gate Program: 112-2926-I-005-504-G), Taiwan; D. K. Wang, P. Izak, W. Kujawski acknowledge the mobility funding by the "Excellence Initiative-Research University" program at Nicolaus Copernicus University in Torun, Poland; and P. Izak, P. Stanovsky acknowledge the support of the Czech Science Foundation, under project 24-10288S, Czech Republic.

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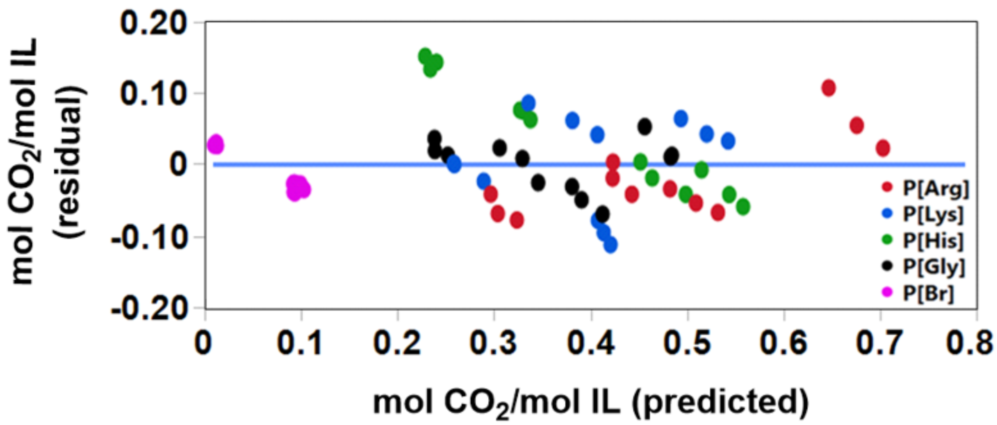


Figure 1. Residual vs predicted CO_2 absorption in AAILs of the FLR model. The blue horizontal line indicates the zero-residual reference (Residual = 0).

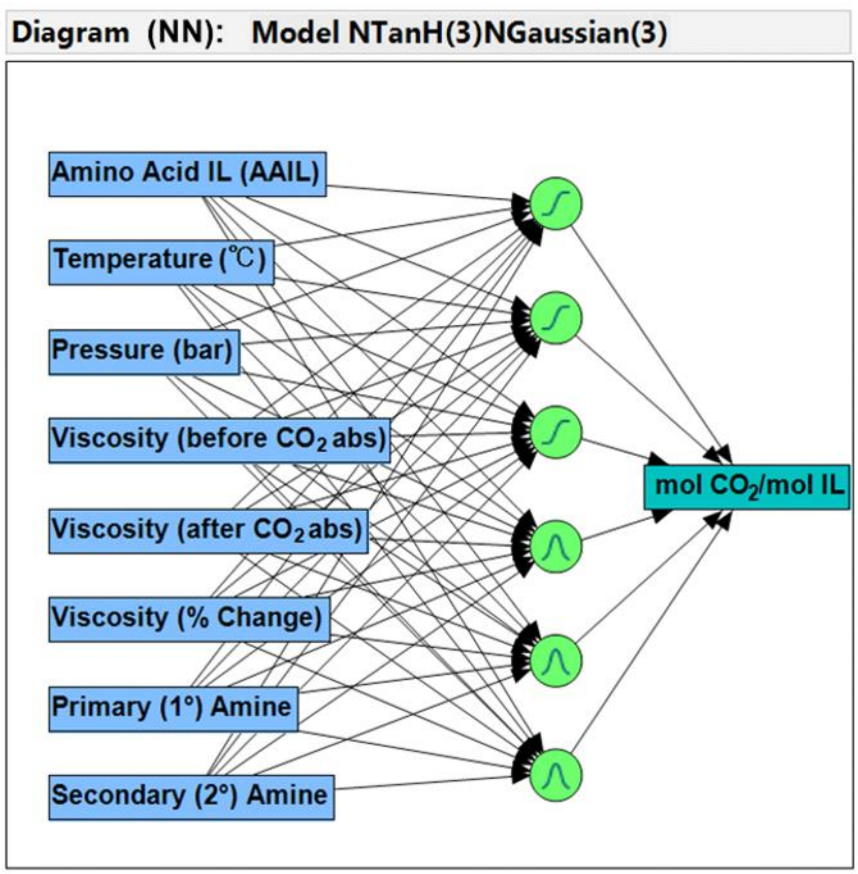


Figure 2. Neural network (NN) diagram for predicting CO_2 absorption in AAILs.

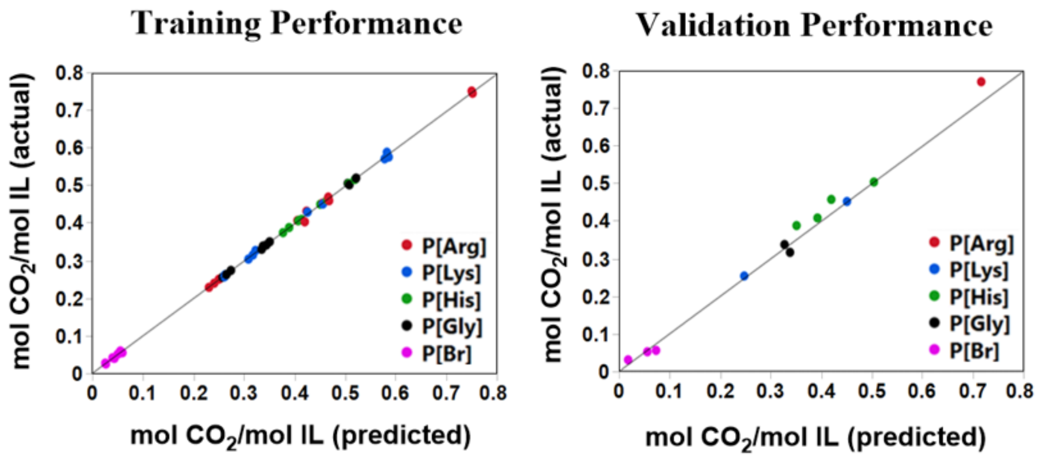


Figure 3. Training and validation performance of the NN model for CO_2 absorption in AAILs.

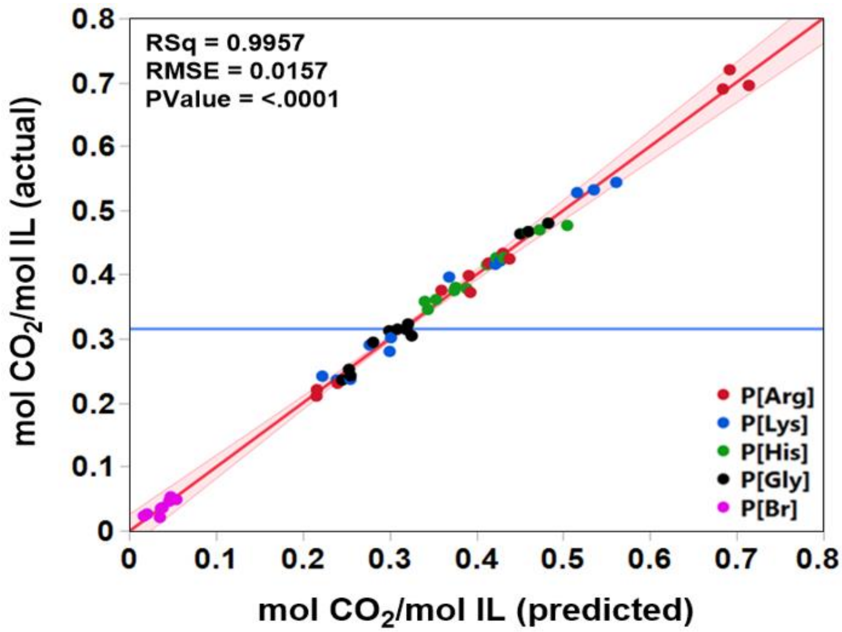
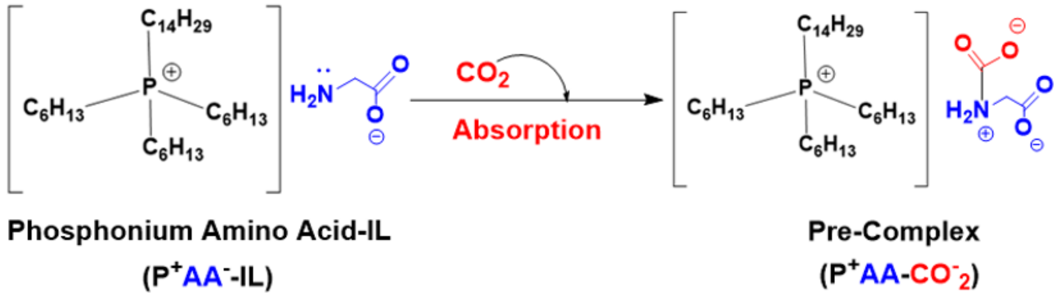


Figure 4. Actual vs predicted CO_2 absorption in AAILs of the SLR model. The red line represents the fitted regression, the shaded red region denotes the 95% confidence interval of the fit, and the blue horizontal line indicates the mean experimental CO_2 absorption (mean response) of the dataset.



Scheme 1. CO_2 absorption mechanism in AAILs.

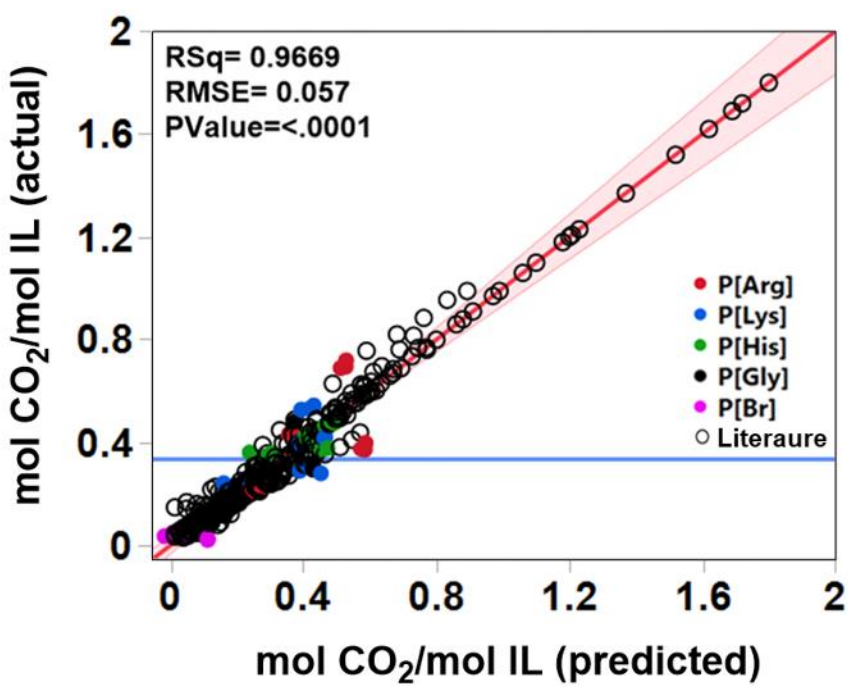


Figure 5. Simulating CO_2 absorption for the experimental and literature-reported AAILs dataset by using the SLR model. The red line represents the fitted regression, the shaded red region denotes the 95% confidence interval of the fit, and the blue horizontal line indicates the mean experimental CO_2 absorption (mean response) of the dataset.

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Table 1. Statistical parameter estimates of the FLR model.

Descriptor	Parameter	Value	Standard Error	t Ratio	Prob > |t|
Intercept	w_0	0.2399086	0.096889	2.48	0.0166*
Temperature (°C)	w_1	-0.003853	0.001883	-2.05	0.0459*
Pressure (bar)	w_2	0.0364779	0.037858	0.96	0.3397
Viscosity (before CO_2 absorption)	w_3	-0.449897	0.336714	-1.34	0.1873
Viscosity (after CO_2 absorption)	w_4	0.3468493	0.354167	0.98	0.3319
Viscosity (% Change)	w_5	0.0016412	0.000615	2.67	0.0101*
Primary (1°) Amine	w_6	0.0276611	0.01813	1.53	0.1331
Secondary (2°) Amine	w_7	0.0121903	0.012007	1.02	0.3147

Note: * indicates statistically significant coefficients at $p < 0.05$ (95% confidence level).

Table 2. Statistical effect test of the FLR model.

Source	Sum of Squares	F Ratio	Prob > F
Temperature (°C)	0.01594986	4.1852	0.0459*
Pressure (bar)	0.00353819	0.9284	0.3397
Viscosity (before CO_2 absorption)	0.00680367	1.7853	0.1873
Viscosity (after CO_2 absorption)	0.00365516	0.9591	0.3319
Viscosity (% Change)	0.02718408	7.1330	0.0101*
Primary (1°) Amine	0.00887129	2.3278	0.1331
Secondary (2°) Amine	0.00392794	1.0307	0.3147

Note: * indicates statistically significant coefficients at $p < 0.05$ (95% confidence level).

Table 3. Training and validation datasets of the NN model.

Training Performance			Validation Performance	
Measures	**Value**		**Measures**	**Value**
RSquare	0.9934743		RSquare	0.9880608
RASE	0.0228132		RASE	0.0221043
Mean Abs Dev	0.0111311		Mean Abs Dev	0.0157447
-LogLikelyhood	-134.6336		-LogLikelyhood	-29.49276
SSE	0.0249812		SSE	0.0058632
Sum Freq	48		Sum Freq	12

Table 4. Statistical effect summary of the SLR model.

Source	Logworth	PValue
Primary (1°) Amine*Secondary (2°) Amine	5.721	0.00000
Temperature (°C)*Viscosity (after CO_2 abs)	4.678	0.00002
Temperature (°C)*Viscosity (% Change)	3.227	0.00059
Temperature (°C)*Viscosity (before CO_2 abs)	2.964	0.00109
Primary (1°) Amine	2.179	0.00662
Viscosity (after CO_2 abs)*Viscosity (% Change)	1.834	0.01464
Viscosity (after CO_2 abs)*Secondary (2°) Amine	1.715	0.01929
Pressure (bar)*Primary (1°) Amine	1.688	0.02052
Temperature (°C)*Primary (1°) Amine	1.594	0.02550
Viscosity (before CO_2 abs)*Primary (1°) Amine	1.498	0.03180
Viscosity (% Change)*Primary (1°) Amine	1.066	0.08591
Pressure (bar)*Viscosity (after CO_2 abs)	0.989	0.10261
Temperature (°C)	0.688	0.20512
Pressure (bar)*Viscosity (before CO_2 abs)	0.542	0.28684
Pressure (bar)	0.536	0.29110
Viscosity (% Change) * Secondary (2°) Amine	0.522	0.30074
Temperature (°C) * Secondary (2°) Amine	0.518	0.30369
Pressure (bar) * Secondary (2°) Amine	0.350	0.44645
Pressure (bar)*Viscosity (% Change)	0.171	0.67377
Secondary (2°) Amine	0.084	0.82507
Temperature (°C)*Pressure (bar)	0.045	0.90172
Viscosity (before CO_2 abs)*Secondary (2°) Amine	0.032	0.92925
Viscosity (after CO_2 abs)*Primary (1°) Amine	0.022	0.95098
Viscosity (before CO_2 abs)*Viscosity (after CO_2 abs)	0.003	0.99295

Table 5. Comparison of developed models with literature-reported models for CO_2 absorption using ionic liquids.

Model Types	Descriptors	Reported Performance	References
FLR (single effects)	Process and functionality (e.g., viscosity changes, T, P, and amine groups) parameters	$R^2 = 0.893$ RMSE = 0.062	This work
NN	Same parameters as above	$R^2 = 0.994$ RASE = 0.023	This work
SLR (two-way interactions)	Curvature and pairwise with parameters	$R^2 = 0.996$ RMSE = 0.016	This work
GNN / fingerprint-ML	Graph/fingerprint	$R^2 = 0.988$, MAE = 0.014	[16]
ANN / LSTM	53 features (incl. T, P, functional groups)	$R^2 = 0.986$ (ANN test), RMSE = 0.027; $R^2 = 0.985$ (LSTM test)	[29, 30]
MLR / ANN / RF	σ-profile (COSMO-RS)	$R^2 = 0.975$, MAE = 0.026 (RF)	[31]
Quadratic/RSM Process vars (e.g., T, P)	Process vars (e.g., T, P)	Polynomial surfaces are widely used; RSM and ML for ILs	[32, 33]
MLR (GA-selected) vs LS-SVM	Structure-based descriptors	Non-linear > linear; MLR as interpretable baseline	[34]

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